

The Cognitive Eye: Indexing Oculomotor Functions for Mental Workload Assessment in Cognition-Aware Systems

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ABSTRACT

Cognitive load assessment has become an integral part of human-centered design as it enables cognition-aware systems to adapt to users' needs. Among the range of physiological metrics, task-evoked pupillary responses have been implemented successfully to determine mental workload. Only recently, smooth pursuit eye-movements as another oculomotor indicator was suggested to index cognitive processing demands. The current study provides a comparison of both parameters with regard to the capacity to differentiate task load in an auditory n-back task. Results replicate earlier findings, suggesting pupil diameter to display the level of mental workload. However, task loads during high levels of difficulty are not distinguishable. Task difficulty also influences gaze behaviour during smooth pursuit eye-movements, although far less sensitive. A clear distinction is achieved only between low and high workloads. In contrast, a combination of control measures (reaction times, accuracy rates) is suited to distinguish between all n-back task stages.

CCS CONCEPTS

• **Human-centered computing - Human computer interaction (HCI) - HCI design and evaluation methods - Laboratory experiments;**

KEYWORDS

Cognitive load, Pupil diameter, Smooth pursuit eye-movements, N-back task

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1 INTRODUCTION

In recent years, cognitive load became a key factor in Human-Computer Interaction (HCI) research. Introduced in the 1980s as a theory of learning (Cognitive Load Theory, [1]), measurement

of cognitive load is nowadays suggested to be an integral part of human-centered design as it can be applied to support work rhythms [3], prevent cognitive overload for purposes of driver safety [4, 5] or guide decisions during interface design.

Cognitive load corresponds to the capacity utilization of the working memory while performing a certain task. In current research, it is estimated by employing a variety of methods. According to Yuksel et al. [6], the most frequently used measurements of cognitive load (in learning studies) are self-reporting questionnaires, followed by dual-task techniques and physiological parameters. The latter primarily include heart-rate variabilities [7, 8] as well as electrodermal activity [9]. However, in the past years, HCI research increasingly applied task-evoked pupillary responses (TEPR) as a quantitative measure to index cognitive processing demands [2, 10, 36, 37]. Such transient pupillary responses reflect changes of mental workload in multiple domains, including (but not limited to) speech perception [11, 12], arithmetical processing [13, 14] or memory performance [15, 16]. Thereby, the majority of studies agree that pupil diameter increases with the level of cognitive load until task difficulty exceeds the available processing resources [2, 17, 18].

Despite its advantages (contactless measurement, high temporal resolution), pupil size changes are particularly sensitive to ambient lighting conditions and cannot be applied in any type of ubiquitous computing system. Latest research deployed another oculomotor parameter to differentiate between levels of cognitive load: smooth pursuit eye-movements (SPEM). SPEM describes a voluntary behavior for the purpose of keeping the retinal image of a moving object within the foveal area. Maintenance of pursuit movements is realized by long tracts that connect frontal brain areas with the cerebellum, whereas performance quality is supposed to be driven by a combination of visual feedback and the prediction of the target velocity [19, 20]. Barnes [21] unanimously emphasizes the decisive influence of vigilance and attention on SPEM; however, he suggests that extra-retinal mechanisms rapidly take over to sustain pursuit movements. Be that as it may, performance accuracy varies during SPEM tasks, whereas errors are corrected by small catch-up saccades to realign the image [22]. Early on, Contreras et al. [20] report cognitive load to influence pursuit eye-movements. Tallying with this, Kosch et al. [23] recently observe mental workload to increase gaze differences during SPEM, especially when following circular and sinusoidal trajectories.

The present work aims to contrast the sensitivity of oculomotor parameters (here: pupil diameter and SPEM) for indexing cognitive processing demands. We apply a dual task scenario as this technique has been proven successful to determine mental workload in the recent past [6]. The level of difficulty is varied on basis of an n-back task. N-back tasks are continuous-recognition measures that comprise stimulus sequences including occasional pairings of

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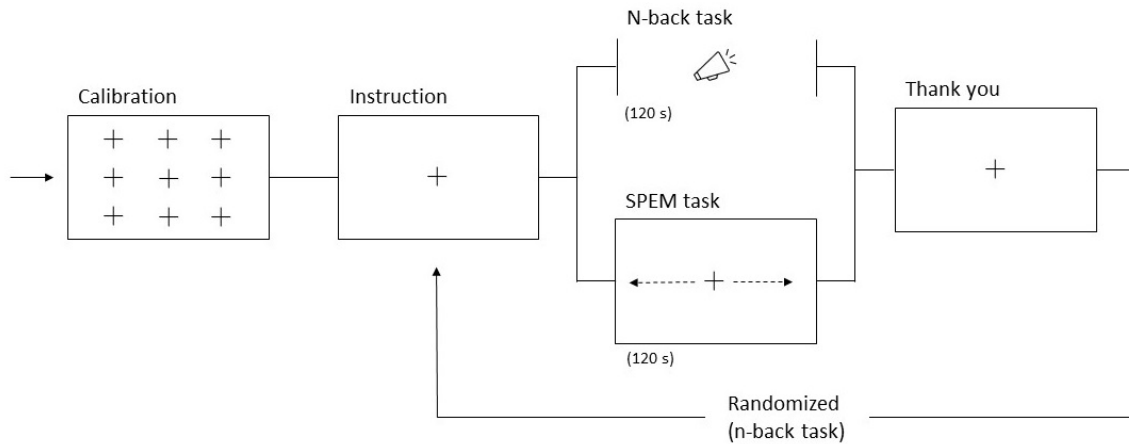


Figure 1: Experimental procedure

numbers or letters with n stimuli in between. The higher the number of elements in between, the higher the level of task difficulty [24]. In order to avoid interferences with the pupillary light reflex, we decided for an auditory n -back task. The impact of task during the n -back task is determined through changes in pupil diameter and on basis of the performance in a simultaneous SPEM task. Conversely, smooth pursuit eye-movements are not assumed to impair the n -back task performance [23, 34]. Behavioural metrics (reaction times, accuracy rates) are applied as control measures. We forgo any questionnaires on perceived taskload as these tools are prone to the influence of individual higher cognitive biases.

2 METHODS

The following subsections provide all relevant information on the experimental design.

2.1 Design and Procedure

The experiment took place in a small laboratory room with an indoor temperature of 21°C and an ambient illumination of 200 lux. Participants were seated in an upright position in front of a video-based eye-tracker that was attached to a monitor at a distance of 90 cm. The right hand was positioned next to a response button in order to press it with minimum effort during task processing. Audio output was realized through a speaker box.

Figure 1 depicts the experimental procedure. Prior to the beginning, a 9-point calibration was carried out to ensure appropriate gaze-tracking. In case of low accuracies (data-target off-set $> 1.5\text{px}$), the calibration procedure was repeated. As indicated below, the dual-task scenario included a cognitive exercise of varying difficulty (n -back task) (2.1.1) and a simultaneous oculomotor assignment (SPEM task) (2.2.2) that remained the same throughout the experiment. We applied a within-subject design. To exclude effects of practice and repetition, participants randomly passed all four trials of the n -back task (control condition, 1-, 2-, 3-back). Before the start of each trial, the task at hand was explained by means of a standardized instruction. Hereafter, an auditory number sequence (n -back task) was provided as a prerecorded file through the speaker box,

while simultaneously a fixation cross moved horizontally across the screen (SPEM task). Participants were instructed to carefully follow the movement with their eyes while processing the n -back task. Each trial took two minutes; short breaks in-between were applied to prevent spill-over effects.

2.1.1 N -back task. Overall task was to continuously recognize recurring numbers that occurred at various positions in the auditory sequence. Task difficulty varied between 1-back (low workload), 2-back (moderate workload) and 3-back trials (high workload). Participants were instructed to press a button as soon as a number was repeated (1-back task) or whenever a number recurred with one (2-back task) or two others in between (3-back task). Feedback on correct or incorrect responses was not provided. Each task comprised four targets. The control condition involved no auditory sequence; participants were instructed to solely process the SPEM task.

2.1.2 Smooth pursuit eye-movements (SPEM) task. A white fixation cross (4 cm high/wide, visual angle 2.6°) was displayed against the black screen background. At the beginning of each trial, it appeared on the left side of the screen and slowly moved towards the right side. Once there, it returned to left side and so on. One crossing of the screen took 15 seconds. Participants were instructed to follow the fixation cross with their eyes while simultaneously processing the auditory n -back task.

2.2 Data Analysis

According to Mantiuk [38], data from the left eye were taken to analyze pupil diameter and smooth pursuit eye-movements. Both parameters were corrected for outliers and missing values via interpolation and multiple imputation. For multiple imputation we created 15 data sets with different imputed values. The MICE algorithm was applied for five iterations leading to one data set without missing values (for details see Wilson [25]).

The first five seconds of each two-minutes pupil diameter data set served as a baseline. Baseline period therefore involves the very beginning of dual task processing. Baseline mean was subtracted

from all subsequent values. Data were smoothed using a one-second moving average. During the 2-back task, we observed occasional data loss at the end of individual data sets, probably because the eye-tracker lost track of the pupillary signal. To obtain equally long sets for diagrams and statistical analyses, all trials were cut to a sample size of 5707 values.

The smooth pursuit eye-movement data were subtracted from 540 (Y-coordinate of the moving cross) and the absolute values of the differences were applied for further processing. A moving average of one second was applied to smooth the data stream. Again, due to data loss during the 2-back task, the data was cut to a length of 5997 values.

2.3 Apparatus

Eye-movements and pupil size changes were recorded with the Tobii Pro Nano featuring a sampling rate of 60Hz. Tobii Pro Lab software (v. 1.123) was applied for data preprocessing and storage. The moving fixation cross was generated with Adobe Premiere Elements and displayed on a 22-inch monitor (Dell P2213) featuring a resolution of 1080x1920 pixels. Instructions were provided via a Bose soundlink box. An URCERI Digital Light Meter was used to measure ambient light conditions. For data analyses and statistics, we used Python (v3.6.5, [26]), the packages NumPy [27], pandas [28], Matplotlib [29], SciPy [30] and miceforest [25] as well as Microsoft® Excel® [31] and the open-source software JASP (v0.14, [32]).

2.4 Sample

27 volunteers (12 female, 15 male, mean age: 26 years, SD: 3, range: 20-35) participated in the current study. All were students of the international master program Human-Computer Interaction at Bauhaus-Universität Weimar and received course credits. They had normal or corrected-to-normal vision and specified no history of head injury and no neurological or psychiatric disorders. Information on regular medication was not collected.

3 RESULTS

The following subsections summarize all relevant findings. The assumption check of normality indicates that our data are not well modelled by a normal distribution. A Friedman test together with Conover's post hoc tests were applied to determine differences in TEPR, SPEM and control measures. P-values during post hoc comparisons are based on the estimation of Bonferroni-corrected values.

3.1 Task-evoked pupillary responses (TEPR)

Pupil diameter increases with the level of n-back task difficulty. Figure 2 depicts pupil size changes as differences from baseline mean and as a function of task load during the n-back task. In control trials, pupil diameter is smallest and decreases steadily while remaining below baseline activity almost the entire time. During 1-back tasks, TEPR display a similar trend, although on a higher level. Pupil diameter occasionally drops below baseline mean halfway through the trial. Two-back and 3-back tasks feature largest pupil diameters of comparable size; a trend to decrease over time can no longer be observed. Across all conditions, TEPR exhibit a restless and fluctuating behavior. However, particularly during

1-back trials (and less obvious during the 2-back and 3-back task), target presentation is followed by a distinct pupil dilation.

Task load during the n-back task exerts a significant influence on pupil diameter ($\chi^2(3)=59.49$, $p<.001$; Kendall's W: 0.60). Pairwise comparisons indicate that the control condition features smaller diameters (M: -0.25 mm, SD: 0.28) compared to the 1-back (M: 0.08 mm, SD: 0.20) ($T=3.15$, $p<.05$), 2-back (M: 0.39 mm, SD: 0.42) ($T=6.19$, $p<.001$) and 3-back task (M: 0.38 mm, SD: 0.23) ($T=6.82$, $p<.001$). Also, pupil diameter differs between the 1-back and 2-back task ($T=3.04$, $p<.05$) as well as between the 1-back and 3-back task ($T=3.67$, $p<.01$). In contrast, we do not observe differences in TEPR between 2-back and 3-back trials.

3.2 Smooth pursuit eye-movements (SPEM)

Deviations in gaze behavior during smooth pursuit movements increase with increasing task load in the n-back task. Figure 3 illustrates raw data from the eye-tracker. Fixations (circles) and saccades (lines) are superimposed and color-coded according to individual participants ($n=27$).

Figure 4 depicts gaze deviations within the SPEM task as a function of task load during the simultaneous n-back assignment. For purposes of presentation, a moving average of three seconds was applied to smooth the data. Similar to changes in pupil diameter, the signal course appears restless and fluctuating. For the 3-back task we observe an increase in gaze deviation over time. Distinct changes linked to target onsets cannot be observed.

Friedman test indicates that the level of difficulty during the n-back trials exerts a significant influence on SPEM ($\chi^2(3)=21.93$, $p<.001$; Kendall's W: 0.69). Pairwise comparisons reveal minor deviations during the control task (M: 32.36 px, SD: 20.64) compared to 2-back (M: 61.52 px, SD: 42.84) ($T=3.57$, $p<.01$) and 3-back trials (M: 66.34 px, SD: 50.84) ($T=4.20$, $p<.001$). In contrast, we do not observe any differences in SPEM between the control condition and the 1-back task. The same is true for comparisons between 1-back and 2-back trials. Contrasting SPEM during the 1-back and 3-back trials reveals a tendency towards stronger deviations within the 3-back task ($T=2.52$, $p=.08$). Finally, we do not observe differences in gaze behavior between 2-back and 3-back trials.

3.3 Control measures

During the n-back task, participants were instructed to press a button as soon as a number was repeated (1-back task) or recurred with one (2-back task) or two others in between (3-back task). Each task comprised four targets. From these data we derived individual reaction times, false-positive responses and omissions. Reaction times were recognized as valid given that a key press occurred within three seconds after a target (onset of the second number). Otherwise, the reaction was considered an omission. Key presses with no connection to a target were counted as false-positive reactions.

Reaction times differ with varying n-back task load ($\chi^2(2)=37.00$, $p<.001$; Kendall's W: 0.43). During 1-back trials, participants respond faster to target appearances (M: 0.73 s, SD: 0.11) compared to 2-back (M: 1.20 s, SD: 0.23) ($T=4.02$, $p<.001$) and 3-back trials (M: 1.50 s, SD: 0.43) ($T=5.96$, $p<.001$). Between the 2-back and 3-back task we do not observe considerable differences.

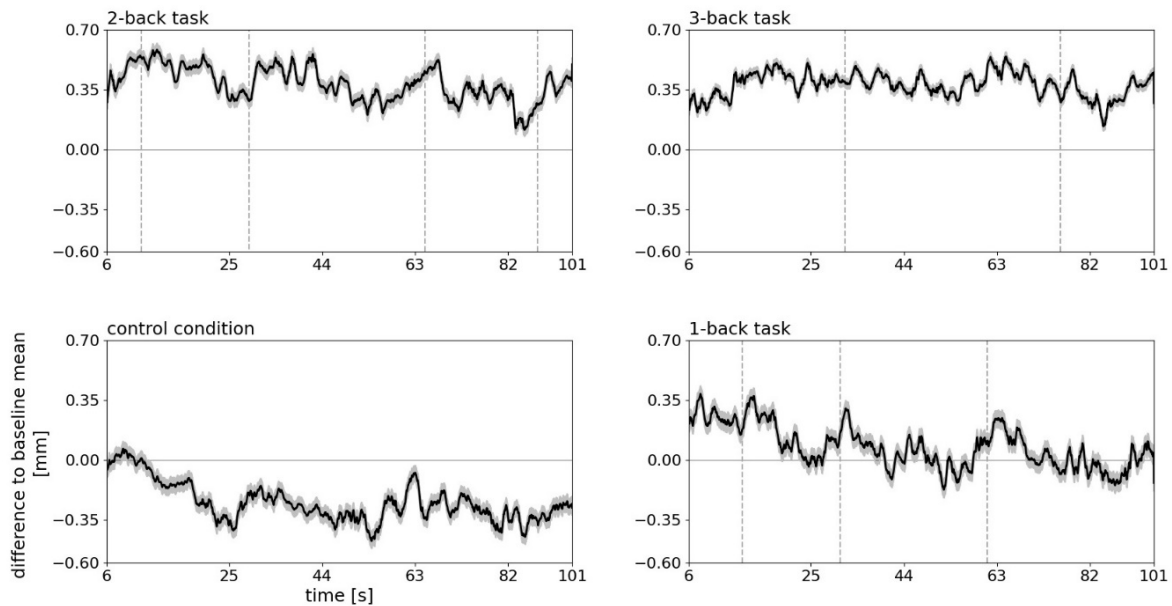


Figure 2: Total averages of pupil size changes (n=27) during the SPEM trials and different levels of the simultaneous auditory n-back task. Values depict differences from baseline mean (set to zero on the y-axis). Darkly colored areas depict 95% confidence intervals. Vertical lines indicate target onset during the n-back task.

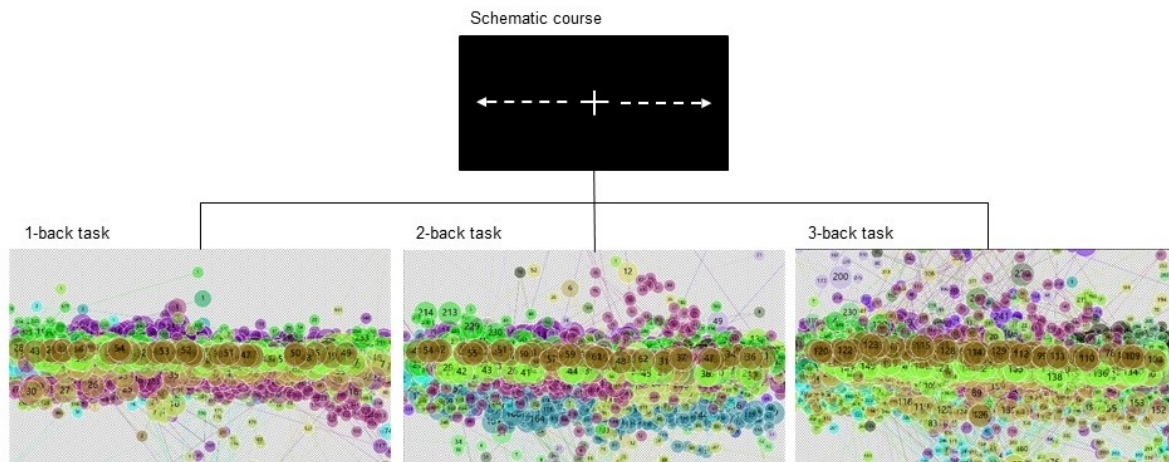


Figure 3: Fixations (circles) and saccades (lines) during the SPEM trials and different levels of the simultaneous auditory n-back task (control condition not depicted). Diameters indicate length of fixations. Individual results (n=27) are superimposed and color-coded.

Task load also influences the number of false-positive reactions ($\chi^2(2) = 40.58$, $p < .001$; Kendall's W : 0.34). One-back trials do not include any false-positives and differ from 2-back (M: 3.86, SD: 0.99) ($T = 6.37$, $p < .001$) and 3-back trials (M: 2.60, SD: 5.53) ($T = 2.75$, $p < .01$). Also, 2-back trials differ from the 3-back task ($T = 3.62$, SD: $p < .001$), whereas 2-back trials involve more false-positive reactions.

Likewise, the number of omissions increases with the level of n-back task difficulty ($\chi^2(2) = 37.54$, $p < .001$; Kendall's W : 0.39).

During 1-back (M: 0.15, SD: 0.36) and 2-back trials (M: 0.11, SD: 0.32), participants overlook a similar number of targets. However, the number of omissions differs between the 1-back and 3-back task (M: 1.56, SD: 1.01) ($T = 5.23$, SD: $p < .001$). Also, 3-back trials involve a higher number of omissions compared to 2-back trials ($T = 5.47$, $p < .001$).

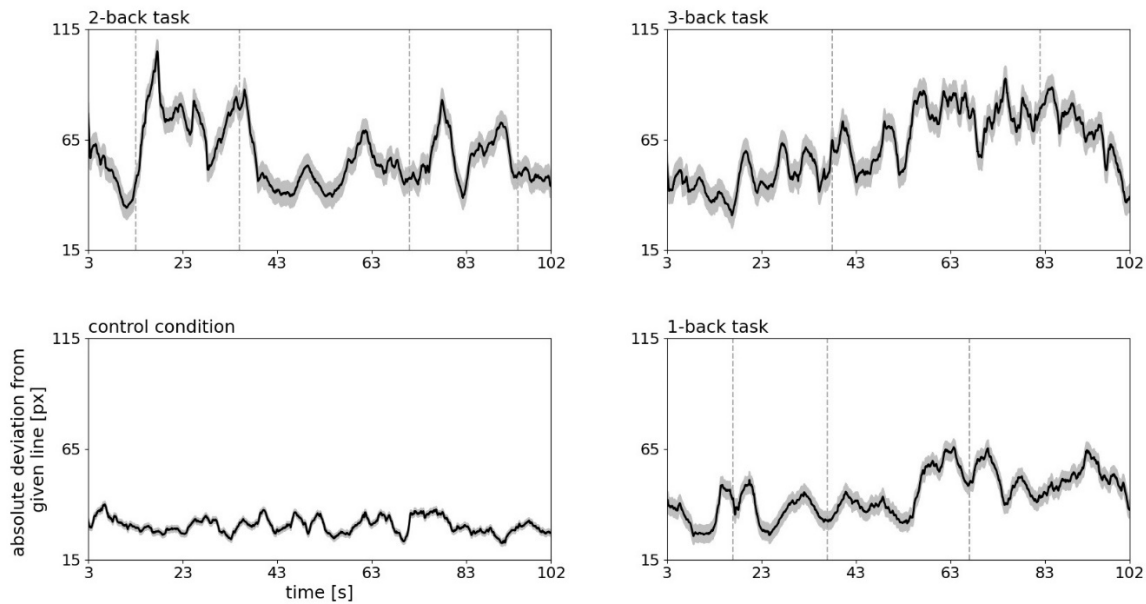


Figure 4: Total averages ($n=27$) of absolute gaze deviations during the SPEM task and different levels of the simultaneous auditory n -back task. Individual gaze points were subtracted from the y -coordinate of the moving target. Darkly colored areas depict 95% confidence intervals. Vertical lines indicate target onset during the n -back task.

4 DISCUSSION AND CONCLUSION

The current study provides a comparison of pupil diameter and smooth pursuit movements with regard to the capacity to index cognitive processing demands in an auditory n -back task. It can be noted that pupil diameter as well as gaze deviations during smooth pursuit increase with increasing task difficulty. However, sensitivity for subtle differences in processing loads challenges the validity of both parameters.

During the control condition and the 1-back task, pupil diameter declines over time. This effect has been documented before [39] and can be attributed to a rather low inhibition of parasympathetic activity. In contrast, higher levels of task difficulty (2-/3-back) involve a considerable increase in pupil diameter (or sympathetic activity) that is capable to inhibit parasympathetic activity. Be that as it may, our findings replicate the influence of cognitive workload on pupil size as reported in earlier studies [2, 10, 14, 33]. Krejtz et al. [2] observe differences in pupil diameter between a control, easy and hard condition. The same is true for Peysakhovich et al. [33] who report pupillary responses to be larger for 1-back compared to 2-back trials. Chen and Epps [10] apply five different levels of task load and obtain effects only among the less difficult trials. This is in accordance with the current findings as we do not observe differences in pupil diameter between 2- and 3-back tasks. It can therefore be assumed that pupil diameter reaches a plateau slightly above moderate task load. Unfortunately, states of high to very high levels of cognitive load are those associated with restrictions in behavioral performances [14]. Accordingly, pupil diameter differentiates between easy to moderate levels of cognitive processing but fails to support the accurate detection of mental overload. Quite the

contrary: According to Zekveld and Kramer [35], pupil diameter starts to decrease at very high levels of cognitive load.

Stubbs et al. [34] report n -back tasks to have negligible effects on horizontal and vertical SPEM. In some ways, the current data are conflicting, given the continuous increase in gaze deviation with increasing level of task difficulty, at least in the descriptive statistics. Stubbs et al. vary the level of difficulty between 1- and 2-back tasks and - in contrast to the current findings -, they do not observe differences in reaction times. Taking this measure as an index of cognitive processing demands, it is conceivable, that there is too little variation in task difficulty for SPEM to be clearly affected. In the present study, the 3-back task reveals a statistical trend to induce stronger deviations compared to the 1-back condition. We therefore assume horizontal SPEM to map differences in cognitive processing demands only for task load conditions that differentiate sufficiently.

Kosch et al. [23] report cognitive workload to increase differences in SPEM. However, authors do not observe an influence while following rectangular trajectories but during circular and sinusoidal movements. Also, they emphasize the speed of the trajectories to exert a considerable effect on the overall SPEM performance. According to Kosch et al., it requires more effort to follow complex movements compared to simple horizontal trajectories. In fact, it is reasonable to assume changes in the allocation of cognitive resources if SPEM becomes faster and more difficult. Processing high levels of the n -back task may not allow to simultaneously carry out challenging SPEM with adequate precision. Complex and fast SPEM should therefore be particularly vulnerable for influences of secondary task load. Furthermore, if n -back task difficulty becomes high enough, it can be expected that even the accuracy of slow and

horizontal SPEM suffers. The current results provide first indications on this issue. Further research should vary both, n-back as well as SPEM task difficulty to determine the level of interaction.

At this point it can be noted that pupil diameter finely differentiates lower states of cognitive load whereas horizontal SPEM is suitable to distinguish extreme states of mental processing. However, given the results in the control measures, it seems advisable to also monitor behavioral performance for a comprehensive assessment of cognitive workload.

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